Incompatible Fairness Criteria in Recidivism Risk Scoring: A Replication of the COMPAS Dispute

Shruti Pingle | August 2026 | Replication study

Executive summary. The 2016 dispute over the COMPAS recidivism score was not a disagreement about facts. ProPublica found that the errors of the score fall disproportionately on Black defendants. Northpointe found that the score is equally predictive across groups. Both findings replicate exactly against the published data. They are compatible because they measure different properties. Where group base rates differ these properties cannot both be satisfied. The observed disparity is therefore not an artefact of model construction and would not be removed by a better model.

1. BACKGROUND

COMPAS assigns criminal defendants a risk score from 1 to 10, which courts use to inform bail and sentencing decisions. In 2016 ProPublica concluded the instrument was biased against Black defendants and the vendor published a rebuttal concluding it was not. The argument that followed treated the two positions as contradictory. This report tests whether they are.

2. DATA AND METHOD

The analysis uses the public ProPublica release of 7,214 defendants screened in Broward County, Florida between 2013 and 2014, with re-arrest outcomes observed over two years. Following the original specification, risk categories are collapsed into a binary prediction in which Medium and High count as flagged. Every figure below was validated cell by cell against the published contingency tables of the source analysis, and that verification is scripted so it fails on any drift. A secondary specification applying the 30 day arrest to screening filter is reported as a robustness check. The measured disparity is 1.9 times under both.

3. FINDINGS

3.1 Error rates are not balanced across groups

As shown in Figure 1, among defendants who did not reoffend, 44.8 percent of Black defendants were flagged high risk compared with 23.5 percent of white defendants. The asymmetry inverts for the complementary error. Among defendants who did reoffend, 47.7 percent of white defendants were classified low risk compared with 28.0 percent of Black defendants.

Each error type falls predominantly on a different group. In a custodial context the two errors do not carry equivalent costs.

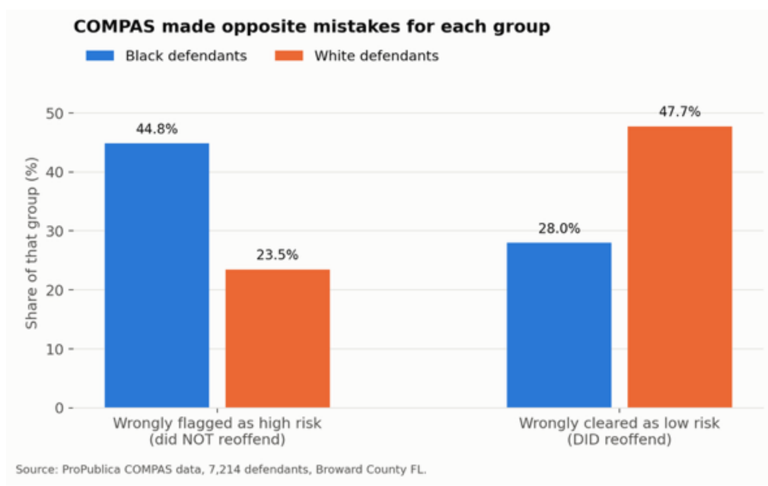

Figure 1. False positive and false negative rates by race. COMPAS risk score, Broward County FL, 2013 to 2014 (n = 7,214). Source: ProPublica.

3.2 Predictive parity holds

Conditional on score, outcomes are close to identical across groups (Figure 2). Positive predictive value is 0.63 for Black defendants and 0.59 for white defendants. Reoffence rates by decile track one another across the full scale.

By the calibration standard advanced in the rebuttal the instrument is defensible. It does not systematically overstate risk for either group.

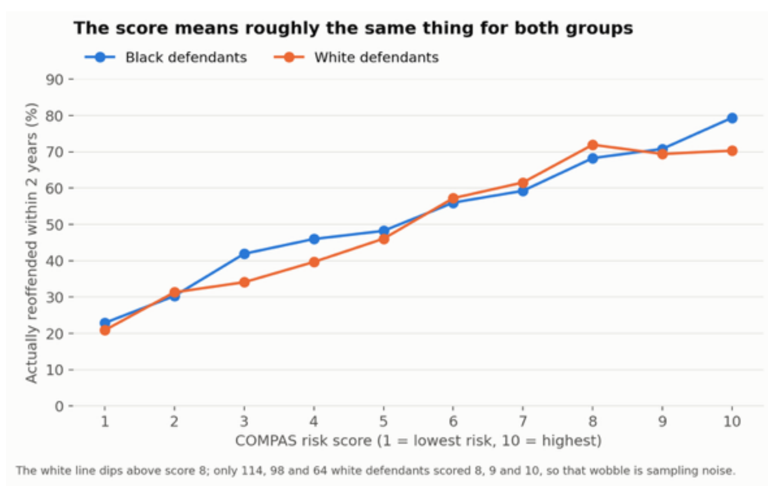

Figure 2. Observed two year reoffence rate by COMPAS decile score and race. Overlapping curves indicate predictive parity. Source: ProPublica.

3.3 The two criteria are mutually exclusive

False positive rate, positive predictive value, false negative rate and prevalence are related by identity:

$$FPR = \left(\frac{p}{1 - p} \right) \times \left(\frac{1 - PPV}{PPV} \right) \times \left(1 - FNR \right)$$

Any three of these quantities determine the fourth. Base rates in this population are 51.4 percent and 39.4 percent. Imposing equal positive predictive value and equal false negative rate across groups therefore still produces a false positive disparity of 1.63 times. In the ratio of the two false positive rates the PPV and FNR terms cancel. What remains is exactly the base rate odds ratio, which is a property of the populations and is invariant to the choice of operating point. This was confirmed across 800 parameter combinations. Figure 3 sets the observed disparity against the disparity a perfectly even handed instrument would still produce.

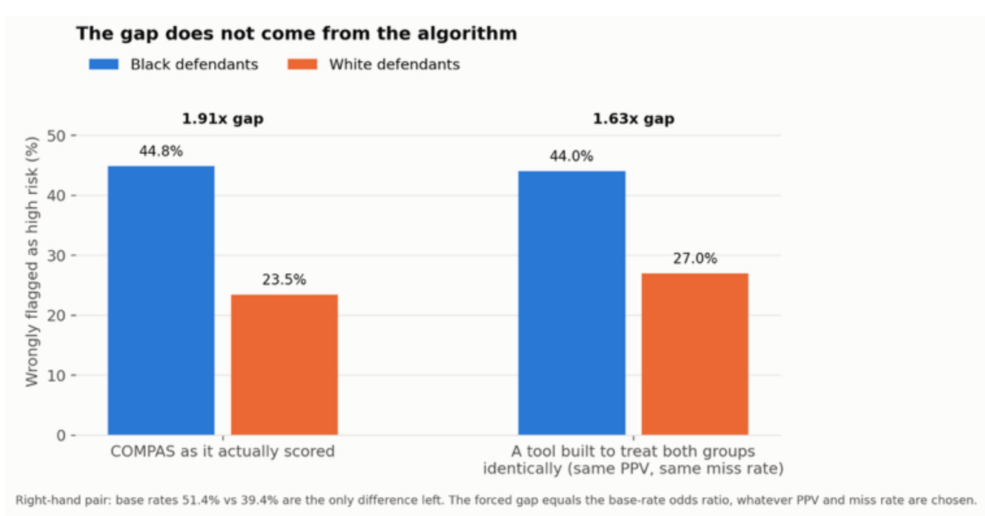

Figure 3. Observed false positive disparity compared with the disparity forced by unequal base rates alone, holding positive predictive value and false negative rate equal across groups.

4. IMPLICATIONS

Fairness is not a single quantity available for optimisation. Equal predictive value and equal error rates are competing definitions. Selecting between them determines which error a system prefers to make and therefore who bears the cost of it. That selection is a policy judgement. Assigning it to model development conceals the decision rather than resolving it, and leaves an accountable choice embedded in a technical artefact.

Limitations. The outcome variable records re-arrest and not offending. Arrest probability is conditioned on enforcement patterns, so the base rate differential is partly endogenous to policing and is not a direct behavioural measure. Calibration is not strictly monotonic for white defendants above decile 8, where sample sizes are 114, 98 and 64. Confidence intervals at those deciles overlap substantially and the reversal is consistent with sampling variation. Findings are specific to one jurisdiction and one period and do not generalise without further work.

References. Angwin J, Larson J, Mattu S, Kirchner L. Machine Bias. ProPublica, 23 May 2016; data at github.com/propublica/compas-analysis. Chouldechova A. Fair prediction with disparate impact. arXiv:1703.00056, 2017. Kleinberg J, Mullainathan S, Raghavan M. Inherent Trade-Offs in the Fair Determination of Risk Scores. ITCS 2017, arXiv:1609.05807. This report reproduces the cited analyses and does not originate them.

Replication code. github.com/shrutipingle02/algorithmic-fairness-audit